Rethinking Utility in Differentially Private Visualization: A Channel-Centric Perspective

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ABSTRACT

Differential privacy (DP) is increasingly adopted to protect sensitive data in visualization, but understanding how privacy-induced noise affects visual representations remains fragmented. Prior work has examined the impact of DP noise on specific chart types, yet these findings are typically tied to particular chart types and evaluation settings, making it difficult to generalize across contexts. In parallel, visualization research has long established that visual channels, the visual properties of marks that encode data attributes, differ in perceptual effectiveness under noise-free conditions. However, these principles have not been systematically revisited under DP, where noise is deliberately introduced into the data.

In this paper, we present a channel-centric perspective to reframe prior empirical and theoretical results. Our synthesis draws on prior findings to motivate a channel-centric hypothesis: differences in the utility of DP visualizations may partially depend on the task-relevant reads made salient by visual channel configurations and DP mechanism choice jointly. Moreover, this perspective highlights the role of channel choice throughout the privacy-preserving visualization pipeline, from disclosure framing and DP algorithm design to utility evaluation and interpretation of noisy outputs. The synthesis also identifies gaps in the existing evaluation literature and highlights research opportunities for channel-aware DP visualization.

Index Terms: Differential privacy, Visual channels, Privacy-preserving visualization, Visualization evaluation.

1 INTRODUCTION

Differential privacy (DP) has become a widely adopted paradigm for protecting sensitive data in statistical analysis and visualization [5]. It ensures that the output of an analysis remains statistically indistinguishable when any single individual's data is added or removed [6]. More formally, given any neighboring databases $\mathscr{D}$ and $\mathscr{D}'$ that differ by at most one record, a randomized mechanism $\mathscr{A}$ satisfies **(ε, δ)-DP** if for any subset $S \subset range(\mathscr{A})$,

$$\Pr[\mathscr{A}(\mathrm{D}) \in \mathrm{S}] \leq \mathrm{e}^{\varepsilon} \cdot \Pr[\mathscr{A}(\mathrm{D}') \in \mathrm{S}] + \delta. \quad (1)$$

The parameters $\varepsilon \geq 0$ and $\delta \geq 0$ specify the degree of privacy provided by the mechanism. Smaller ε and δ values indicate stronger privacy protection, and vice versa. For aggregate queries, DP guarantees can be achieved by adding random noise into query results, such as through Laplace mechanism and Gaussian mechanism [6]. The noise magnitude depends on the query's sensitivity and privacy parameters (smaller ε and δ indicate stronger privacy and may yield larger noise). Most existing studies on DP visualization [10, 19, 28] adopt the ε-DP model (i.e., setting $\delta = 0$). They also adopt algorithmic approaches such as data aggregation and transformation to preserve utility while providing DP guarantees.

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Building on Saket et al.'s [23] task-based evaluation of chart effectiveness under noise-free conditions, Zhang et al. [28] revisited the same chart-task combinations under DP noise. They evaluated four chart types across eight tasks from the Amar et al. [1] taxonomy and found that *summary tasks* (e.g., Find Clusters) were more noise-tolerant than *value tasks* (e.g., Retrieve Value). They also observed that pie charts offered similar or better accuracy than bar charts under noise, a result that diverges from clean-data chart rankings. However, they characterized which task–visualization combinations tolerated noise, but did not examine *why* certain chart types gained or lost advantage. Subsequent work in DP visualization has expanded the design space [10, 15, 19, 21], further motivating a need to connect these findings across studies and examine them through a common lens.

Visualization research has long established that visual channels (e.g., position, length, angle, and color/intensity), the perceptual dimensions used to encode data attributes into visual marks [16], differ in perceptual effectiveness [4]. Because chart types are constructed from combinations of visual channels, a channel-centric perspective provides a natural lens for connecting findings across chart types and reasoning about why utility differs under DP. However, this perspective has not yet been used to synthesize findings across DP visualization studies.

In this paper, we synthesize prior empirical and theoretical work on DP visualization [3, 10, 15, 18, 19, 20, 21, 22, 25, 26, 27, 28, 30] through a channel-centric lens. We identified relevant work using Zhang et al. [28], Panavas et al.'s study [19], and the survey of Bhattacharjee et al. [2] as starting points for backward and forward citation snowballing, supplemented by independent keyword searches (e.g., *differential privacy*, *visual utility*). We prioritized studies examining how visualization affects utility under DP.

Our synthesis yields three main insights that become apparent when prior studies are considered together. First, chart-specific design choices, such as bin resolution, component allocation, and rendering geometry, influence utility under DP. Second, initial evidence of chart-task ranking shifts under DP motivates a channel-centric hypothesis: DP visualization utility may partially depend on the types of task-relevant reads made salient by visual channel configurations and DP mechanism choice jointly. Third, channel choice plays a role across the DP-visualization pipeline, from disclosure framing and DP algorithm design to utility evaluation and interpretation.

This paper makes three contributions: (1) a channel-centric synthesis of prior empirical and theoretical work on DP visualization; (2) a hypothesis that task-relevant reads made salient by visual channel configurations and DP mechanism choice jointly impact visualization utility; and (3) gaps and research opportunities for channel-aware evaluation and design in DP visualization.

2 EVALUATION CONTEXT: A UTILITY HIERARCHY

Evaluating the utility of privacy-preserving visualizations is itself a multi-faceted problem. Bhattacharjee et al. [2] identify a fundamental lack of consensus on how to quantify this utility. We characterize three complementary evaluation lenses used in prior work, organized along a hierarchy from low-cost automated methods to

high-cost human evaluation.

Visual utility measures how closely a private visualization preserves features relevant for downstream tasks, typically through automated metrics: image-similarity measures or task-specific distribution scores. Panavas et al. [19] benchmarked automated metrics against expert-perceived visual quality for differentially private binned scatterplots; MS-SSIM (multi-scale structural similarity) emerged as the strongest predictor, while APQE (average per-query error) and Earth Mover's Distance (EMD), two standard DP utility metrics, correlated negatively with perceived visual quality. Ryan et al. [22] showed through user studies that Pixel Approximate Entropy (PAE) correlates with perceived visual complexity in line charts, though not under DP noise. Zhang et al. [28] illustrate the latter category, introducing Peakedness, Anomaly, and Clusteredness scores tailored to three analytical tasks (Find Maximum, Find Anomalies, Cluster) to compare DP algorithms by how well their noisy output preserves the distribution features each task requires.

Task-based utility measures whether human viewers can correctly perform analytical tasks on noisy charts. Zhang et al. [28] proposed a dichotomous assessment model that cleanly separates perceptual accuracy (did the user correctly read the noisy chart?) from perturbation accuracy (does the noisy answer match the true answer?), enabling targeted diagnosis of where utility breaks down. Task-based evaluation is the most direct measure of utility but also the most expensive, requiring human participants for each combination of conditions.

Perceptual utility operates along a distinct axis: it measures whether an authorized viewer can read the chart while unauthorized viewers cannot, addressing physical observation threats rather than inference from released outputs [3, 29].

Notably, the automated metrics used at the lower end of this hierarchy already differ by chart type and associated channel: MS-SSIM for binned scatterplots [19], PAE for line charts [22]. Whether this pattern reflects a deeper need for channel-specific evaluation is a question we return to in Section 4.

3 A Channel-Centric Synthesis of DP Visualization Utility

We organize our cross-study synthesis into three themes. Each draws on multiple studies whose connections become apparent through the lens of visual channels.

3.1 Within-Chart Design Choices Influence DP Utility

Even for a single chart type, DP utility is not determined by noise magnitude alone. Design choices such as DP mechanism/composition, bin resolution, color scale, and aspect ratio of the rendering can substantially shape utility at a given ε. We illustrate this through discretization parameters, component sensitivity, and rendering geometry.

Discretization parameters. Panavas et al. showed that, holding the privacy level and DP algorithm constant, coarser bins (32×32) retain significantly more visual utility than finer bins (64×64) [19]. Ghazi et al. [10] suggested a possible explanation: finer grids split the same data across more bins, reducing per-bin signal while leaving DP noise magnitude largely unchanged. This lowers the signal-to-noise ratio of spatial patterns, suggesting that the bin-size effect reflects a broader discretization challenge in DP spatial visualization.

Component sensitivity. Ramsay and Diaz-Rodriguez [21] showed that even within a single boxplot, whose components all use length/position channels, the same DP mechanism performs on different components unequally, where the box represents the median and quartiles and the whiskers depend on extreme quantiles. Their composite DPBoxplot uses different DP estimators for different components, showing that component-aware design can reduce utility loss relative to a uniform approach.

Rendering geometry. Ryan et al. [22] showed that line chart geometry, independent of the underlying data, substantially shifts visual complexity, as measured by PAE: doubling chart width reduces PAE by more than half, while doubling height increases it more than twofold. Combined with their experimental finding that shape identification accuracy drops sharply once PAE exceeds a threshold, this implies that aspect ratio may mediate utility for private line charts: a wider rendering may lower perceived visual complexity, helping keep the chart below the threshold at which trends become unrecoverable. Thaker et al. [27] made a complementary observation for bar/histogram length encodings: at sufficiently large data scales, DP perturbations can fall below the screen's pixel resolution, making a noisy bar visually indistinguishable from its non-private counterpart. Together with Ryan et al.'s PAE result, this shows that rendering geometry can act as a within-chart design lever.

3.2 Chart-Task Observations under DP

In Table 1, we summarize initial evidence of ranking shifts by comparing Zhang et al.'s [28] task-based DP study with Saket et al.'s [23] non-private baseline, supplemented by Panavas et al.'s [19] study of DP binned scatterplots. We caution that each observation derives primarily from a small number of task-based studies under certain DP algorithms and privacy parameter ε, so these patterns should be considered as initial empirical evidence under limited context/conditions.

From a channel-centric perspective, we map the chart-task pairs to task-relevant channel configurations and task-relevant reads in Table 1. Note that charts are often multi-channel encodings and we list the task-relevant channel configuration following canonical channel vocabulary [4, 16]. Piechart is represented by arc length, central angle, and segment area jointly, because prior studies suggest that all three cues can affect proportional reading [7, 24].

We use *read* to refer to the task-relevant quantity a viewer seeks to recover from a visualization. In Table 1, we annotate each chart-task pair with our interpretation of the task-relevant read most directly associated with the corresponding channel configuration. We hypothesize that the observed DP ranking shifts partly reflect interactions among the channel configuration, the task-relevant read it makes salient, and the DP mechanism used. Section 4.1 develops this hypothesis.

Existing work also offers several partial explanations, each developed for a specific channel or task setting. For the lineplot shift (*position*, trend/shape), Ryan et al.'s [22] Pixel Approximate Entropy (PAE) provides a possible mechanism: shape identification accuracy collapses from near-perfect at low PAE to near-chance at high PAE with a nonlinear threshold, and DP noise can raise a lineplot's PAE by introducing discontinuities into the contour. For the higher robustness of binned-scatterplot clustering than distribution characterization (*color/intensity*), Panavas et al. [19] attribute the gap to signal-to-noise: dense regions produce strong color signals that DP noise does not overshadow, while subtle density patterns require fine discrimination that noise disrupts.

3.3 Channel Considerations Across the DP-Visualization Pipeline

Beyond predicting which encoding may be superior for a given task, the channel-centric view shapes practice upstream and downstream of encoding selection. We trace its implications at four points along the DP-visualization pipeline: how disclosure is framed in design, how DP mechanisms are tuned to preserve specific structures, how utility is measured, and how analysts interpret noisy output. At each point, the channel acts not as a passive rendering decision but as an active determinant of what privacy-preserving visualization can deliver.

Table 1: [ADDED] Chart-task ranking shifts under DP, based on prior work [23, 28, 19], with our interpretation of the task-relevant channel configuration and read. Blue ↑: more DP-robust; red ↓: more noise-sensitive.

Chart / Task	Non-private baseline	DP finding	Task-specific channel configuration	Interpreted task-relevant read
Lineplot / Characterize Distribution	Significantly *most* accurate [23].	↓ **Significantly *worse* than other chart types under Laplace + post-processing,** $\varepsilon \in \{0.01, 0.001\}$ **[28].**	Position of connected line marks, forming a trend/shape contour.	Trend / Contour-shape.
Barplot / Determine Range	No significant chart-type differences [23].	↓ **Accuracy degraded significantly as ε decreased under Laplace + post-processing,** $\varepsilon \in \{0.01, 0.001\}$ **[28].**	Length / position on a common scale.	Absolute magnitude / extrema.
Piechart / Retrieve Value	Among the top performers, with no reliable difference from Bar or Table [23].	↑ **Significantly higher perceptual accuracy than other chart types under Laplace + post-processing,** $\varepsilon \in \{0.01, 0.001\}$ **[28], with label/color caveats†.**	Sector geometry: arc length, central angle, and segment area jointly [7, 24].	Proportional / part-whole.
Scatterplot / Filter; Cluster	No significant chart-type differences for Filter and barplot strongest for Cluster [23].	↑ **Significantly superior for both Filter and Cluster under Laplace + post-processing,** $\varepsilon \in \{0.01, 0.001\}$ **[28].**	Two-dimensional position of point marks.	Spatial-layout / grouping.
Binned scatterplot / Cluster vs. Distribution	N/A	↑ **Cluster retained the highest visual utility** and ↓ **Distribution the lowest** under data-independent and -dependent DP histograms, $\varepsilon \in \{0.5, 0.1, 0.05, 0.01\}$ [19].	Color/intensity over spatial bins.	Aggregate / Density-region.

†Saket et al. [23] note that pie charts "may have been further helped by having colored slices with text labels."

Disclosure and channel as paired design choices. Mehta et al. [15] contributed a theoretical framework that treats channel as design-relevant in privacy-preserving visualization. They distinguished *disclosure tactics* (data operations that change what information is available) from *emphasis tactics* (rendering choices like mark types, scales, and layouts). Their Fred scenario illustrates that disclosure tactics and channels are not chosen independently: a biostatistician sharing patient clusters while hiding individuals deliberately pairs the *smoothing* tactic with *contour shape*, because that combination preserves cluster visibility while suppressing individual-level detail. Mehta et al. acknowledged that emphasis-tactic choices "tend to be concomitant to choices about disclosure" [15] but left emphasis tactics out of their analytical scope. Our channel-centric synthesis lays an empirical foundation for that choice under DP noise: Section 3.1 and 3.2 show that channel-related choices, both within and across encodings, may have channel-specific consequences for which utility is preserved.

DP algorithm, channel, and task as joint design choices. Ghazi et al. [10] designed a DP algorithm specifically for heatmap visualization, with EMD as the loss function the algorithm targets, chosen for its spatial-proximity properties on color/intensity-encoded heatmaps. This illustrates DP algorithm design being tailored to a target visualization rather than treated as an independent, upstream choice. Panavas et al. [19] observed a related coupling within a single channel: DAWA [13], a data-dependent algorithm that groups nearby cells into near-uniform regions, reinforces the dense-region signal that cluster identification depends on while smoothing away the subtle density variations that distribution characterization requires. The same algorithm aligns with one task on the color/intensity channel and works against another. Whether the optimal DP algorithm depends systematically on the visual encoding, on the analytical task, or on their combination, and whether algorithms tuned for one such pairing transfer to others, remains an empirical conjecture; no study has directly compared DP algorithms across channels and tasks under matched conditions. The evidence so far suggests that selecting a DP algorithm without considering both the rendering channel and the analytical task risks discarding what the visualization is meant to communicate.

Evaluation metrics need empirical validation, not just theoretical channel-fit. The choice of evaluation metric is itself a design decision that interacts with channels. Ghazi et al. [10] chose EMD for DP heatmap evaluation based on its spatial-proximity properties, a theoretical argument for channel-fitness. Panavas et al. [19] empirically benchmarked five metrics including EMD against expert visual utility ratings on heatmap-style binned scatterplots; MS-SSIM correlated most strongly with human perception, while EMD correlated negatively. Theoretical arguments for a metric's channel-fitness do not automatically translate to predictive power for perceived utility. Panavas et al.'s [20] practitioner interviews confirmed the operational consequence: practitioners reported that general utility metrics are easy to compute but not interpretable for evaluating visualization quality. Validating which channel-specific metric actually tracks human perception requires empirical work that has not been done for most visualization types.

The interpretation bottleneck. Nanayakkara et al. [18] found that analysts using DP outputs lost roughly nine times as much performance to misreading noisy results as to allocating the privacy budget poorly. ViP [17] illustrated the leverage of an encoding choice: it discretizes the continuous Laplace noise distribution into a quantile dotplot, turning probability-mass reading into counting dots rather than estimating area under a curve. A controlled study with 16 practitioners showed analysts read CDF probabilities more accurately from the dotplot than from a numerical-error baseline. This substitution favors count-based judgment over continuous-magnitude reading. If interpretation is a primary bottleneck, encoding choices are where direct leverage lies.

4 Discussion

4.1 A Hypothesis: Task-Relevant Reads Affect Utility in Privacy-Preserving Visualization

Table 1 motivates the hypothesis that the utility of a privacy-preserving visualization partially depends on the task-relevant read made salient by the visual channel configuration. DP mechanisms preserve some properties of the data more faithfully than others. Therefore, the utility of visualization partially depends on how much the adopted privacy mechanism preserves task-relevant read. For example, adopting the Laplace mechanism [6] to achieve DP

does not guarantee the preservation of ordering between groups, thereby may change the upward or downward trend in a lineplot. Prior research shows that post-processing the results of the Laplace mechanism (e.g., to enforce non-negativity) may introduce bias and different error variances among bins [9], which may explain the difficulty with retrieving magnitudes accurately in DP barplots. When large group or dense region read are task-relevant, the Laplace mechanism may be more applicable, as demonstrated in prior work on frequent itemset mining and web mining [14, 12, 8].

This explanation also applies at the within-chart level design and to a broad range of privacy mechanisms. Coarsening bin resolution or performing hierarchical decomposition [10, 19, 27] could better preserve aggregate density and thus retain better visual utility. Ramsay and Diaz-Rodriguez's [21] boxplot decomposition shows that private quantile estimators may perform differently towards inner quantiles and extreme quantiles. Incorporating specific pattern constraints in the DP mechanism may improve the visualization utility in pattern retention [30].

Bhattacharjee et al. [2] earlier proposed a related distinction between high-accuracy channels (position, length) and low-accuracy channels (area, density) in privacy-preserving visualization, framed around *disclosure risk*: high-accuracy channels reveal individuals too precisely and must be transformed into aggregate encodings to hide them. Our hypothesis addresses a different question. Rather than asking which channels disclose more information, we ask how channel configurations make particular task-relevant reads salient and whether the adopted DP mechanism preserves the information required for those reads.

However, existing evidence is limited. No study has isolated the channel-read-mechanism interaction from confounds such as label design, visual complexity, privacy level, or task familiarity. Saket et al. [23] themselves noted that pie charts may have benefited from colored slices with text labels. This hypothesis is based on available results and may lead to testable predictions, but it remains a conjecture awaiting controlled studies.

4.2 Gaps

Visual utility metrics are channel-specific and largely unvalidated. As noted in Section 2, the automated metrics already in use differ by encoding: MS-SSIM for scatterplots [19], PAE for line charts [22]. Other channels, including angle, length, and slope, lack perceptually validated automated metrics. Theoretical arguments for a metric's channel-fitness do not automatically translate to predictive power for perceived utility, as Panavas et al. [19] demonstrate empirically (Section 3). Validating which channel-specific metric tracks human perception requires empirical work that has not been done for most visualization types.

Task-based evidence rests on narrow conditions. Zhang et al.'s [28] task-based user study, the primary source of channel-task evidence under DP, varied chart type, task, and privacy level, but held fixed three dimensions our synthesis identifies as consequential: within-encoding design parameters such as bin resolution and chart geometry, data properties such as scale and distribution, and the choice of DP algorithm (Laplace only). Whether the documented ranking shifts hold under different conditions is unknown. Length encoding has also been studied at the statistical level by Hay et al. [11] across scale, distribution, domain size, and DP algorithm; whether their statistical findings translate into task or visual utility for bar charts remains uncharacterized. Non-tabular visualization types (networks, treemaps, parallel coordinates) are almost entirely unexamined under DP, with Tao et al.'s [26] brain connectivity work a rare exception.

Perceptual utility has been studied largely independently of DP. BAIT [3], a visual-illusion-based perceptual privacy approach, demonstrated uniformly high protection across chart types for shoulder surfing but found channels differ in vulnerability: angle and slope encodings are most "glanceable." Whether channel rankings derived under perceptual privacy align with or diverge from rankings under DP noise has not been examined, leaving the channel-level relationship between the two privacy regimes open.

4.3 Opportunities

A channel-specific metrics suite. The methodology Panavas et al. [19] used to validate MS-SSIM (collecting expert visual utility ratings on a corpus of DP charts, then benchmarking candidate automated metrics against those ratings) may transfer directly to other channels. Developing such a suite would let practitioners compare encodings at scale during design, rather than after deployment through expensive user studies.

Cross-channel comparison at matched ε. Extending Zhang et al.'s [28] design to cover channels underrepresented in the task-based DP literature, and to vary data scale and distribution alongside encoding choice, would provide the first systematic channel evaluation under DP. Controlled user studies could further test whether the channel effectiveness pattern is explained by the relative-vs-absolute distinction (Section 4.1).

Cross-channel evaluation of structure-aware DP algorithms. Panavas et al. [19] showed that DP algorithm choice substantially affects visual utility within a single channel, and that a single algorithm's fit can vary by task on the same channel (Section 3). Whether such differences extend across channels and tasks, and whether algorithms purpose-built for specific channels or tasks (e.g., [10, 21, 25]) transfer to others, has not been tested under matched conditions.

Multi-channel interactions. Real-world encodings rarely depend on a single channel. Visualizations beyond basic charts, such as treemaps, parallel coordinates, and node-link diagrams, are inherently multi-channel. How channels interact under DP and whether noise on one channel compounds or compensates for noise on another are largely open. Such interactions may also complicate algorithm selection: an algorithm optimal for a chart's primary channel may degrade secondary channels. It is possible that DP algorithm design for multi-channel encodings requires joint optimization across channels.

Channel behavior across privacy threat models. Whether BAIT's [3] channel-level vulnerability findings interact with DP-induced degradation is unexplored. A study that varies both observation threat (BAIT-style) and DP noise across channels would clarify whether the two privacy regimes reinforce or counteract each other at the encoding level.

4.4 Limitations

The channel ranking shifts trace to Zhang et al.'s [28] task-based study and are specific to its chart types, tasks, algorithm/parameters, and datasets. The candidate explanation we offer, while consistent with available evidence , has not been tested in a controlled experiment that isolates encoding properties from confounds. Our framework addresses basic visual channels; compound encodings introduce interactions we do not address. Extending to multivariate visualizations, where multiple noisy variables interact, is a significant open challenge [2, 28].

Despite these limitations, our synthesis reveals patterns that are less visible when individual studies are considered in isolation. It shifts attention from "how much noise can we add?" to "how should we encode the noisy data, and how does that choice shape disclosure framing, algorithm selection, and interpretation?" This is a question visualization researchers are uniquely positioned to answer. Panavas et al.'s [20] practitioner interviews reveal that practitioners select chart types for different pipeline stages without an established framework; their interviewees express a desire for empirically grounded design guidance. We hope this work motivates the systematic, channel-aware research that such guidance requires.

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